

Empirical Strategy

Research Title:

Estimating the Causal Impact of the GENIUS Act on U.S. Stablecoin Market Dynamics Using Bayesian Structural Time-Series Modeling.

Research question:

To what extent did the 2025 GENIUS Act alter the relative demand dynamics for U.S. dollar-pegged stablecoins, after accounting for macroeconomic conditions?

Empirical Strategy:

1. Statement of Identification Problem

In this study, we are using the CausalImpact method to investigate the causal effect of The Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS Act) on the stablecoins growth using the combined market capitalization of the top two most used ones as my metric of measurement.

For context, GENIUS Act was signed into law on July 18, 2025, to give regulatory framework for using the stablecoin payment system. It is intended to boost issuers and users' confidence and institutional adoption.

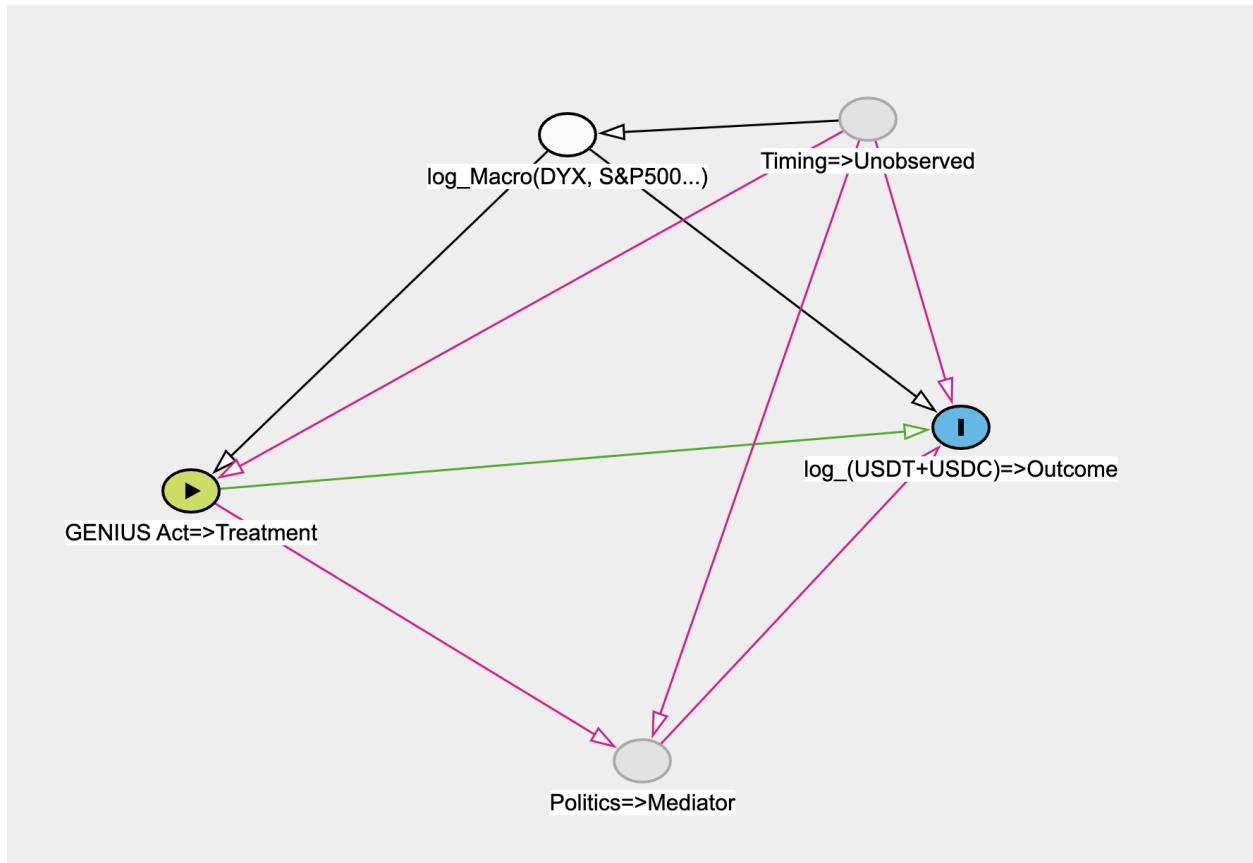
The main challenge for estimating the causal impact is to control for other factors. Stablecoin market caps are affected by several other factors including their organic growth into the wider cryptocurrency market as well as to economic factors such as the dollar, interest rates, equities and other market volatility and events such as cryptocurrency price surges in other countries (e.g. due to MiCA in the EU). A simple pre-post difference-in-differences (DID) would also not be unaffected by the underlying factors that could look like the effect of the Act on stablecoin issuers (confidence) or customers (compliance and demand for stablecoins). Pre-legislative market discussions and associated post-legislative market volatility also needs to be controlled for. We therefore undertake a quasi-experimental analysis using a set of control variables that are not affected by the GENIUS Act. Using Bayesian Structural Time-Series (BSTS) modelling we then construct a valid counterfactual.

2. A Clear DAG (Directed Acyclic Graph)

The DAG (directed acyclic graph) above illustrates the causal pathways and assumptions that we need to identify our strategy. In the DAG, each node is a variable, and each edge represents the causal relationship between variables. Remember that to be a DAG the graph must be acyclic. Nodes: Treatment: GENIUS Act (binary intervention at $t = \text{July 18, 2025}$). Outcome: Combined USDC + USDT Market Cap (Y_t , daily log-transformed). Control/Confounding Variables: Macro indicators ($X_t = [\log(\text{DXY}), \log(10Y \text{ Treasury Yield}), \log(\text{S\&P 500}), \log(\text{VIX})]$). Unobserved Confounders: U_t (e.g. global crypto sentiment, DeFi trends, issuer-specific events like mint/burn decisions). Time-Varying Trends: Local level/trend (μ_t) and seasonality (S_t). Edges: GENIUS Act $\rightarrow$ Outcome: Direct causal effect (policy clarity boosts demand/issuance).

Potential Confounders (X_t) $\rightarrow$ Outcome: Stablecoin demand is affected by macro conditions such as safe haven flows when the VIX is higher. Confounders (X_t) $\rightarrow$ GENIUS Act: Macro factors have an impact on the GENIUS Act. (e.g. Strengthen Dollar) Unobserved: Residual (U_t), and Timet (μ_t) $\rightarrow$ Outcome: Residual drivers of growth and Stochastic trend that captures organic evolution. Backdoor paths from Treatment to Outcome via Confounders are blocked by conditioning on X_t and controlling for unobserved (U_t) effect and time effect (μ_t).

Representation:



3. Key Model Equations

The CausalImpact BSTS model breaks down the outcome into structural components. It then uses knowledge from the prior period to infer what the post-period counterfactual would have looked like.

Key equations (in log-scale for scale/stationarity):

Observation Equation (core model):

$$\mathbf{y}_t = \boldsymbol{\mu}_t + \boldsymbol{\beta}' \mathbf{x}_t + s_t + \epsilon_t, \epsilon_t \sim N(\mathbf{0}, \sigma_\epsilon^2)$$

where:

y_t : log (Combined USDC + USDT Market Cap) at time t.

μ_t : Local level (stochastic trend), evolving as $\boldsymbol{\mu}_t = \boldsymbol{\mu}_{t-1} + \boldsymbol{\delta}_t$, $\boldsymbol{\delta}_t \sim N(\mathbf{0}, \sigma_\delta^2)$ (random walk).

$\boldsymbol{\beta}' \mathbf{x}_t$: Regression on log-transformed macro controls

$\mathbf{x}_t$ (vector: log (DXY), log (DGS10), log (SP500), log (VIXCLS)).

$\boldsymbol{\beta}$ selected via spike-and-slab prior.

s_t : Optional seasonality (e.g., weekly cycle: harmonic terms or dummy).

ϵ_t : Irregular noise.

Counterfactual Prediction (post-intervention):

$$\hat{\mathbf{y}}_t^{cf} = \hat{\boldsymbol{\mu}}_t + \hat{\boldsymbol{\beta}}' \mathbf{x}_t + \hat{s}_t$$

(posterior from pre-period MCMC; no treatment indicator).

Causal Effect:

$$\tau_t = \mathbf{y}_t - \hat{\mathbf{y}}_t^{cf}, \bar{\tau} = \frac{1}{T_{post}} \sum_{t \in post} \tau_t$$

(pointwise and average effects, with credible intervals).

I model the series Bayesianly (with MCMC, 1000+ draws) on the trimmed pre-period (Jan 1, 2024–Apr 17, 2025) and then forecast it into the post period (Oct 19, 2025–Mar 5, 2026).

4. Core Assumptions

Pre-Period Correlation Checks - Stabilizes the Pre-Post Relationship, which verifies that the relationships between the controls ($\mathbf{X}_t$) and the outcome ($\mathbf{Y}_t$) before the intervention are the same after the intervention and that the Act had no unrelated structural breaks.

Exclusion Restriction (ER) for Controls Unaffected by Treatment The four Macros (DXY, yields, S&P 500, VIX) are not affected by the GENIUS Act — a US-focused policy that we can hardly imagine having a spillover to the financial markets macro variables we are aware of.

No Anticipation Effects: We remove the no anticipation effects for this section, primarily to prevent market participants from reacting to the policy before it is implemented. Specifically, we exclude the 3 months leading up to the effective date of the policy (April 18–July 17, 2025) to also minimize the general legislative volatility. We believe it is unlikely for the market to be prepared for the implementation of a law that does not carry any significant economic impact to stakeholders (stablecoins do not have price movements).

Sufficient Pre-Data and Identifiability: Variable Explanatory Models require enough pre-data and identifiability. The long pre-period is about 15.5 months and provides a good amount of trend and seasonality to estimate the underlying trends.

Ignorable Unobservable: After conditioning on X_t and μ_t , residuals (ε_t) are independent and identically distributed (iid) and therefore not correlated with treatment (parallel trends in latent space).

5. Threats to Identification and How to Address Them

Threat: Unobserved Confounders/Time-Varying Bias. The crypto market may experience surprise factors that are not accounted for in the macro and can act as a source of upward bias, examples of this could be the recent rise of the DeFi sector, the BTC halving etc..

Address: These are tests for the placebo of the fake pre-period intervention (e.g. “New Year” for Jan 1, 2025). The goal is to measure and subtract out any organic trend bias. These are some notes on sensitivity analysis for crypto proxies (e.g. the ETH cap) in case the placebo test ends up not working.

Threat: Anticipation or Immediate Volatility: Pre-enactment leaks/speculation or post-signing overreactions contaminate periods.

Address: Includes explicit 3-month exclusion buffers; sensitivity analyses varying window sizes (e.g. 2 months versus 4 months)

Threat: Spillover to other Control measures. Should Act have an indirect positive impact on equities (for example, through a general positive mood in the fintech sector) and exclusion was violated (which in this case would imply an underestimation of the Act’s impact).

Address: Validate with post-period Chow tests for structural breaks in β and drop the suspect controls such as the S&P 500 in the robust specification.

Threat: Measurement Error/Noise: Crypto data volatility or macro forward-filling introduces noise.

Address: Log-transformation for stationarity; weekly aggregation as sensitivity check.

6. Planned Robustness Checks

Placebo Tests:

Use the model to calculate the effects of the fake interventions that we add in the pre-period (e.g. Jan 1, 2025; July 1, 2024) to ensure that there is no real effect and that there is no spurious detection. This will allow us to remove the placebo bias from the real estimate. Change the

length of the time window for the calculations by shortening/lengthening the pre-period (for example 12 months instead of 18 months) and/or decreasing the size of the exclusion buffers (for example 2 months) to see how sensitive the volatility model is.

Secondary Outcome: Repeat on combined USDC+USDT volume (on-chain transfers) — consistent positive effects strengthen usage-dynamics inference.

Secondary Outcome: Repeat on combined USDC+USDT volume (on-chain transfers) — the more times this result holds, the more we can infer what usage-dynamics look like.